

A Story of Numbers — Full Test

Mixed questions from all three concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes ·
Total: 28 marks. Answers on the separate key.

Objective (1 mark each)

Q1 • MCQ**[1]**

The Roman numeral L stands for:

- (a) 5 (b) 40 (c) 50 (d) 500

Q2 • MCQ**[1]**

The number 49 in Roman numerals is:

- (a) IL (b) XLIX (c) XXXXIX (d) LIX

Q3 • MCQ**[1]**

In a base-n system, the landmark numbers are:

- (a) multiples of n (b) powers of n (c) factors of n (d) prime numbers

Q4 • MCQ**[1]**

The place value of the digit 7 in 4736 is:

- (a) 7 (b) 70 (c) 700 (d) 7000

Q5 • FILL IN THE BLANK**[1]**

The number 143 written in base 5 is _____.

Q6 • TRUE / FALSE**[1]**

The Roman number system has no symbol for zero.

True False

Q7 · MCQ

[1]

The Egyptian number system is a:

(a) base-10 system (b) base-5 system (c) base-60 system (d) base-2 system

Q8 · ASSERTION-REASON

[1]

Assertion (A): Zero is essential in the Hindu number system.

Reason (R): Zero acts as a placeholder, so 52 and 502 are different numbers.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

Write 2367 in Roman numerals, showing how you group the number.

Q10

[2]

Convert 324 (base 10) into base 5.

Q11

[2]

Write 47561 in expanded form using powers of 10.

Q12

[2]

Write the place value of the digit 8 in 38420.

Long answer (3 marks each)

Q13

[3]

Convert the base-5 number 1033 into base 10, showing the place values.

Q14

[3]

Convert to Roman numerals, showing your grouping: (i) 444 (ii) 1984.

Q15

[3]

Find the place value and face value of the digit 6 in 36174, and their difference.

Q16

[3]

State three features of the Hindu number system that make it more efficient than the Roman system.

Total: 28 marks · Check your work against the **answer key**.